from the hypothalamus of the brain to all the organs that control the body metabolism, growth and development, reproduction and also maintain homeostasis.

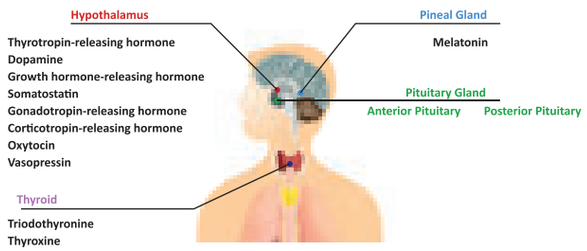
The endocrine system regulates its hormones through negative feedback. Increases in hormones activity decrease the production of that hormone

The endocrine system consists of these glands

- Hypothalamus
- Pineal gland
- Parathyroid
- Pancreas
- Testes
- Pituitary gland
- Thyroid
- Adrenal glands
- Ovaries

NOTE: The Thymus is the only endocrine gland that does not secrete any hormones

The Thymus is considered as an endocrine gland because it plays a role in the immune system of the body.



Endocrine System

Hypothalamus

The hypothalamus lies within the cranium which regulates the homeostasis. The hypothalamus regulates the release or slow down and stop the release of hormones from other endocrine glands based on the blood levels of these hormones.

Pituitary Gland

Located at the base of the brain, is the **master gland** which controls the secretion of several other glands included in the endocrine system.

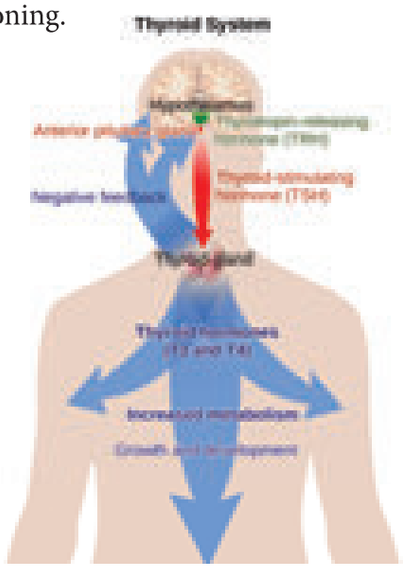
Pineal Gland

The pineal gland is a small endocrine gland that lies close to the hypothalamus, performs the coordination of **circadian rhythm (sleep - wake cycle)**.

Thyroid Gland

The Thyroid gland shaped like angel wings on each side of the throat just above the trachea, are connected to each other with a thin connecting area called the Isthmus.

The Thyroid gland regulates the body's metabolism, basal metabolic rate, cardiac system's function, physical growth and sexual functioning.



Thyroid System

Parathyroid Glands ("para" means around and "thyroid" is thyroid gland).

The parathyroid glands are two pairs of glands found bilaterally on both sides of the neck just behind the Thyroid gland. The body has four parathyroid glands which is unique when compared to the other glands of the endocrine system. The role of the glands is to control the circulating amount of two electrolytes, calcium and phosphorous with the secretion of parathyroid hormone.

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